

WEST PLAINS, MO, USA

WMO: 723484

Lat: 36.878N

Lon: 91.903W

Elev: 373

StdP: 96.93

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53901

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.4	-9.7	-18.8	0.7	-9.6	-16.4	0.9	-7.9	9.9	4.0	8.9	4.2	3.0	330	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	34.8	23.7	33.0	23.6	31.9	23.4	25.5	31.8	24.9	30.9	24.2	29.7	3.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.7	19.4	28.9	23.0	18.5	27.9	22.5	18.0	27.4	80.4	31.7	77.5	30.8	74.9	30.0	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.5	6.8	-16.4	36.7	2.3	2.5	-18.0	38.5	-19.3	40.0	-20.6	41.4	-22.3	43.3		
(5)			-17.0	26.5	2.1	0.7	-18.5	27.0	-19.7	27.4	-20.9	27.8	-22.4	28.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	1.5	3.4	8.9	14.3	18.8	23.5	25.5	25.0	20.8	14.6	8.5	3.0		
	DBStd	9.69	6.19	6.22	5.84	4.68	4.14	2.92	2.81	2.94	3.78	4.85	5.22	5.48		
	HDD10.0	904	271	199	91	16	1	0	0	0	14	89	225			
	HDD18.3	2345	522	418	295	133	47	2	0	1	18	137	297	476		
	CDD10.0	2376	7	14	57	145	274	403	480	464	326	155	43	8		
	CDD18.3	774	0	0	3	13	61	156	222	207	93	19	1	0		
	CDH23.3	7333	0	2	24	115	459	1444	2255	2013	853	159	8	0		
	CDH26.7	2861	0	0	2	14	97	522	1006	898	293	28	0	0		
(14) Wind	WSAvg	3.0	3.5	3.5	3.7	3.7	3.0	2.5	2.2	2.1	2.2	2.7	3.2	3.3		
(15) Precipitation	PrecAvg	1164	76	76	111	126	131	94	99	74	110	94	107	80		
	PrecMax	1422	141	188	349	242	303	167	179	150	309	304	223	173		
	PrecMin	952	11	16	25	30	40	44	39	14	4	6	21	5		
	PrecStd	163	36	40	73	56	56	31	43	35	68	68	63	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.7	22.6	25.8	28.3	31.0	34.8	37.1	37.4	33.4	29.3	24.7	19.0		
		MCWB	14.2	15.3	16.1	18.3	21.2	23.1	23.9	23.8	22.4	20.9	17.2	14.8		
	2%	DB	17.1	19.0	23.3	26.3	29.0	32.5	34.6	35.0	31.9	27.2	21.8	16.7		
		MCWB	13.2	13.1	15.6	17.3	21.0	23.4	24.0	23.6	22.6	19.7	15.1	13.6		
	5%	DB	14.0	16.2	20.9	24.4	27.5	31.1	32.9	33.0	30.1	25.0	19.3	13.9		
		MCWB	10.1	11.8	13.9	16.4	20.4	22.9	24.1	23.5	21.9	18.1	14.2	10.5		
	10%	DB	11.1	13.6	18.3	22.6	26.2	29.7	31.8	31.4	28.0	22.8	17.4	11.6		
		MCWB	7.3	9.5	12.6	15.7	19.8	22.5	23.9	23.2	20.6	17.1	13.2	8.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.2	16.8	18.3	20.3	23.5	25.6	26.4	26.1	24.3	22.7	18.9	16.6		
		MCDB	18.2	21.2	23.7	25.7	28.4	31.6	33.3	32.5	30.2	27.5	22.5	18.2		
	2%	WB	14.3	15.0	16.8	19.2	22.4	24.7	25.7	25.3	23.5	21.4	17.1	14.0		
		MCDB	16.1	17.8	21.4	23.9	27.1	30.3	32.1	31.8	29.0	25.3	19.5	16.0		
	5%	WB	10.8	12.8	15.5	18.1	21.6	23.9	25.2	24.6	22.8	19.9	15.5	11.5		
		MCDB	13.3	15.3	19.2	22.1	25.9	29.2	31.1	30.8	28.0	23.0	18.5	13.1		
	10%	WB	7.5	9.7	13.9	17.0	20.6	23.2	24.5	23.9	22.0	18.4	13.4	9.1		
		MCDB	10.6	13.4	17.8	21.0	24.6	28.1	30.0	29.6	26.7	21.8	16.7	11.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	10.8	11.7	11.9	11.0	10.8	10.8	11.4	12.2	11.9	11.2	9.9		
		MCDBR	14.0	14.4	14.3	13.7	12.2	12.2	12.2	13.6	13.3	13.3	13.7	12.4		
	5% WB	MCWBR	9.9	9.7	8.1	7.1	6.0	5.0	4.4	4.6	5.4	6.9	8.5	9.0		
		MCWBR	12.0	12.6	11.9	11.3	10.3	10.4	10.8	11.4	11.0	11.0	11.3	10.6		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.305	0.337	0.377	0.409	0.423	0.454	0.449	0.385	0.343	0.313	0.296		
	taud		2.487	2.469	2.426	2.317	2.294	2.270	2.230	2.223	2.385	2.487	2.519	2.513		
	Ebn at Noon		895	935	929	904	876	860	829	826	866	879	873	877		
	Edh at Noon		81	93	107	126	132	135	139	137	111	91	78	74		
(44) All-Sky Solar Radiation	RadAvg		2.36	2.96	4.01	5.12	5.67	6.48	6.25	5.65	4.85	3.61	2.55	2.00		
	RadStd		0.22	0.36	0.30	0.33	0.50	0.44	0.36	0.33	0.34	0.41	0.28	0.24		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+134
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page